

10(a)	<i>any seven from:</i> 1 impulse / action potential / depolarisation, reaches, synaptic knob ; 2 (voltage-gated) Ca^{2+} channels open (in pre-synaptic membrane) ; 3 Ca^{2+} enters (synaptic knob / pre-synaptic neurone) ; 4 vesicles with acetylcholine, move towards / fuse with, <u>pre-synaptic membrane</u> ; 5 (ACh) released / secreted / exocytosis ; 6 ACh <u>diffuses</u> across <u>synaptic cleft</u> ; 7 binds to receptors on post-synaptic membrane ; 8 (ligand-gated) Na^{+} channels open and Na^{+} enters post-synaptic neurone ; 9 depolarisation / action potential / EPSP ; 10 <u>acetylcholinesterase</u>, breaks down ACh / recycles Ach ;	7
-------	--	---

Question	Answer	Marks
10(b)	any eight from: 1 Hill reaction ; 2 oxidised DCPIP is blue ; 3 (DCPIP) goes colourless when reduced ; 4 method for making chloroplast, extract / suspension ; 5 <i>ref. to</i> buffer solution / control osmosis / control pH ; 6 expose chloroplasts + DCPIP to wavelength of light ; 7 (method 1) measure time for, blue to disappear / blue-green → green / colour to match control with no DCPIP or (method 2) leave for fixed time and measure with colorimeter ; 8 calculate rate as, (method 1) $1/t$ or (method 2) change in colour value divided by time ; 9 test five (or more), wavelengths / colours ; 10 do three repeats / calculate means ; 11 plot wavelength on x-axis against, dependent variable / calculated rate, on y-axis ; 12 AVP ;	8